E9 246 Advanced Image Processing
Project Proposal

March 6, 2026

Dwaipayan Haldar, Bitupan Arandhara

Project Topic and Problem Statement

Title: Analysis and Improvement of TokenCut for Unsupervised Object Discovery.

Problem Statement: Object segmentation is a fundamental problem in computer vision. Most modern approaches rely heavily on supervised learning and require large amounts of annotated data. However, unsupervised object discovery aims to identify the main object in an image without any human-provided labels. This is particularly useful in scenarios where labeled datasets are scarce or expensive to obtain.

Recently, Wang et al. proposed *TokenCut* [1], an unsupervised object discovery method that combines self-supervised Vision Transformer (ViT) features with spectral graph partitioning. The method constructs an affinity graph between image patches using feature similarity and then applies the Normalized Cut algorithm to separate foreground and background regions.

Although TokenCut demonstrates strong performance compared to previous unsupervised methods, it relies on several assumptions about the structure of the image and the connectivity of the affinity graph. In many realistic situations, these assumptions fail, leading to incorrect segmentation results.

The goal of this project is to analyze the limitations of the TokenCut framework and explore potential improvements to make the method more robust in challenging scenarios such as occlusion, multiple objects, and large foreground regions.

Team Members

- Dwaipayan Haldar, M.Tech Signal Processing
  SR No.: 04-03-01-10-42-25-1-27128
- Bitupan Arandhara, M.Tech Research
  SR No.: 04-03-00-10-22-25-1-25910

Project Objectives

- **Understanding TokenCut:** Study the theoretical foundation of TokenCut including Vision Transformer features, construction of affinity graphs, and the normalized cut formulation used for spectral graph partitioning.
- **Implementation:** Implement the TokenCut pipeline including feature extraction, affinity matrix construction, eigenvalue decomposition, and foreground segmentation.
- **Identify Limitations:** Experiment with different images to identify scenarios where TokenCut fails. In particular, we will focus on three major limitations:
  - Occlusion of objects
  - Presence of multiple objects
  - Large foreground objects occupying significant portions of the image
- **Explore Improvements:** Investigate possible modifications to improve segmentation performance under these challenging scenarios.

Methodology

1. Baseline Implementation

We will first implement the TokenCut pipeline as described in [1]. The key steps include:

1. Extract patch-level features from a Vision Transformer model trained using self-supervised learning (e.g., DINO).
2. Construct an affinity matrix based on cosine similarity between patch features.
3. Apply thresholding to sparsify the affinity graph.
4. Solve the normalized cut eigenvalue problem to obtain the second smallest eigenvector.
5. Perform bipartition based on the eigenvector values to obtain the foreground mask.
6. Apply post-processing to refine the segmentation mask.

2. Analysis of Limitations

We will evaluate the behavior of the algorithm on a variety of images and study the structure of the affinity matrix and segmentation outputs in different scenarios.

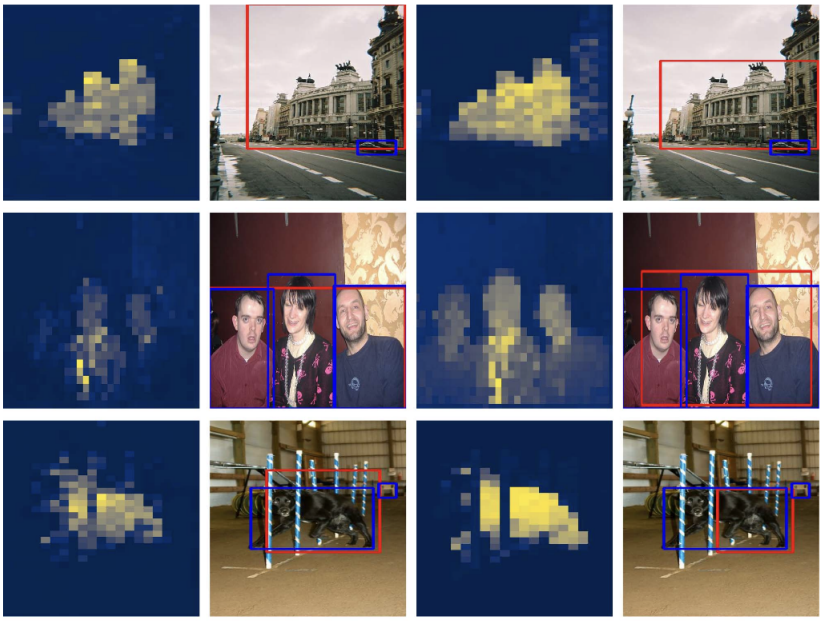
The following failure cases will be investigated:

Occlusion: When an object is partially occluded by another object or structure, the similarity graph between object patches may become fragmented, causing the spectral partition

to split the object incorrectly.

Multiple Objects: TokenCut assumes the presence of a single dominant foreground object. In images containing multiple objects, the algorithm may detect only one object or merge multiple objects into a single cluster.

Large Objects: TokenCut focuses on the largest salient part in the image, which may not be the desired object.



(a) LOST Inverse Attn. (b) LOST Detection (c) Our Eigen Attention (d) Our Detection

Fig. 8. *Failure cases on VOC12 (1st and 2nd row) and COCO (3rd row).* LOST [16] mainly relies on the map of inverse degrees (a) to perform detection (b). For our approach, we illustrate the eigenvector in (c) and our detection in (d). Blue and Red bounding boxes indicate the ground-truth and the predicted bounding boxes respectively.

Figure 1: Above is the excerpt from the paper, that shows the failure case

3. Proposed Improvements

Based on the observed limitations, we will explore the following possible improvements:

1. **Multiple Object Detection**

   (a) **Multi-eigenvector clustering:** TokenCut uses only the second smallest eigenvector of the normalized cut problem. We will compute multiple eigenvectors

$y_2, y_3, \ldots, y_k$ to embed patches into a spectral space, then apply k-means clustering. Each cluster can correspond to a distinct object, generalizing the binary partition to a multi-way segmentation.

(b) **Iterative TokenCut (recursive graph partition):** Run TokenCut repeatedly: detect and remove object patches, then re-run TokenCut on the remaining graph. This allows sequential discovery of multiple objects while keeping the core algorithm intact.

(c) **Adaptive thresholding on the eigenvector:** Instead of splitting at the mean value of the second eigenvector, detect multiple peaks in the eigenvector distribution. Each peak corresponds to an object seed, which is then grown into a region.

(d) **Graph community detection:** Replace the normalized cut bipartition with community detection algorithms such as the Louvain method or modularity maximization on the affinity graph. These methods naturally discover multiple clusters without a fixed number of partitions.

2. **Occlusion Handling**

(a) **Spatial continuity constraint:** Modify the affinity matrix to combine feature similarity with a spatial proximity term. This ensures nearby patches remain connected in the graph even when their appearance differs due to partial occlusion.

(b) **Multi-hop graph connectivity (graph diffusion):** Instead of relying solely on direct patch similarity E_{ij}, incorporate two-hop connections via graph diffusion:

$$E' = E + \gamma E^2$$

This reconnects object parts separated by an occluder through intermediate patches.

(c) **Multi-eigenvector segmentation with spatial merging:** Occluded objects often appear as two disconnected clusters in the second eigenvector. Using additional eigenvectors y_2, y_3 and merging spatially proximate clusters helps recover the complete object region.

(d) **Region merging post-processing:** After running TokenCut, identify multiple candidate foreground clusters and merge those that are both spatially close and feature-similar. This serves as a post-processing step that compensates for graph fragmentation caused by occlusion.

4. Evaluation

We will evaluate the performance of the baseline TokenCut implementation and the proposed modifications using both qualitative and quantitative analysis.

Evaluation will include:

- Visual inspection of segmentation masks
- Intersection-over-Union (IoU) comparison with ground truth masks
- Analysis of affinity matrix structures in successful and failed cases

References

References

[1] X. Wang, et al., "TokenCut: Segmenting Objects in Images and Videos with Self-Supervised Transformer and Normalized Cut," *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

[2] M. Caron et al., "Emerging Properties in Self-Supervised Vision Transformers," *ICCV*, 2021.

[3] J. Shi and J. Malik, "Normalized Cuts and Image Segmentation," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2000.